

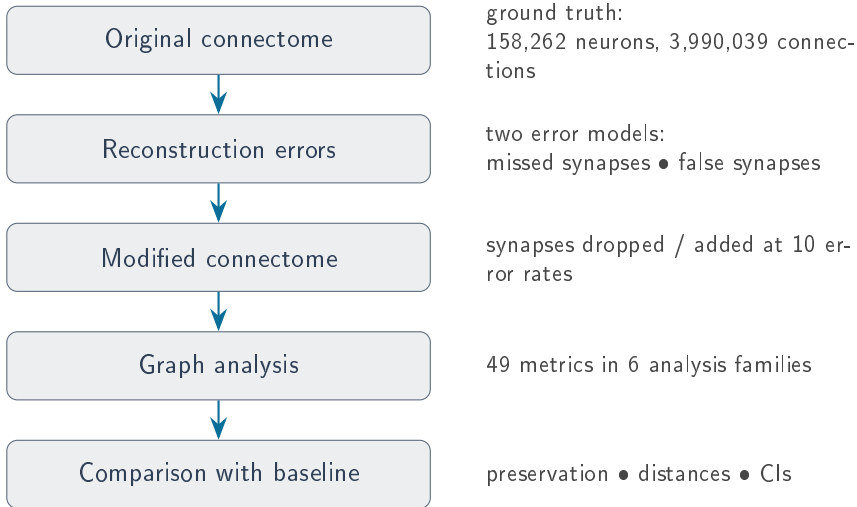
Impact of Reconstruction Errors on Connectome Graph Analysis

Experimental Analysis of Missed Synapses and False Synapses

BANC Dataset

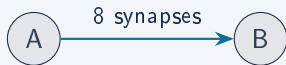
Neuron- and synapse-resolved connectome simulation study · 10 error rates · 5 trials each

Problem Overview



Two Error Models

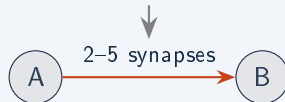
Missed synapses (false negatives)



edge removed if
weight reaches 0

Each synapse survives independently (binomial);
weight = number of surviving synapses.

False synapses (false positives)



new edge appears

$k = \text{rate} \times \text{edges}$ artificial connections added;
weight sampled from the weak-edge distribution
($\text{syn} \leq 5$).

Experimental Setup

Dataset	BANC — <i>D. melanogaster</i> brain + nerve cord
Baseline	158,262 neurons · 3,990,039 connections 23,556,214 synapses
Error rates	0, 0.25, 0.5, 0.75, 1, 2, 5, 10, 15, 20%
Trials	5 seeded per rate (10 rates × 2 models)
Metrics	49 metrics in 6 analysis families
Runtime	≈3 min per trial (≈5 h total)

Pipeline (fixed order)

Load → Build → Preprocess →
Perturb (error model) →
Analyze → Statistics → Export

Analysis families

structure · degree distributions ·
SCC/WCC · reciprocity · assortativity ·
PageRank · similarity distances

Evaluation Metrics — Grouped by Graph Property



Topology

edge count
density
degree distributions



Connectivity

weak / strong
connected
components
(count, giant size)



Synaptic properties

total synapses
weight mean
/ median
variance / max



Network organization

reciprocity
assortativity
PageRank

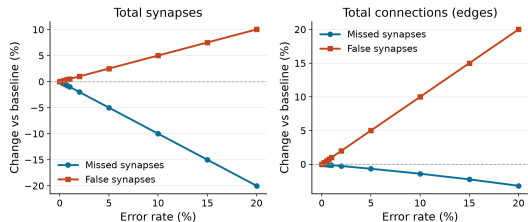


Similarity

KS / Wasserstein
Pearson /
Spearman
top-*k* overlap

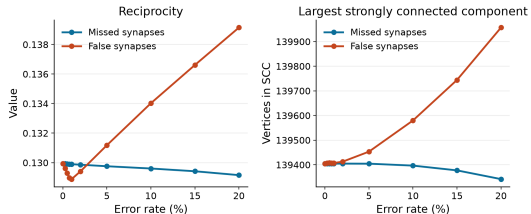
49 metrics in five categories, compared with baseline (95% CIs, 5 trials).

Missed Synapses — Proportional Erosion, Intact Topology



Synapses: loss is exactly proportional to rate (−20.0% at 20%).

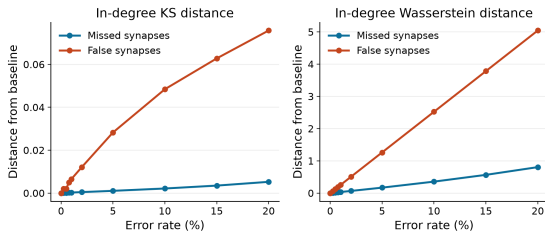
Edges: only −3.2% — connections rarely vanish.



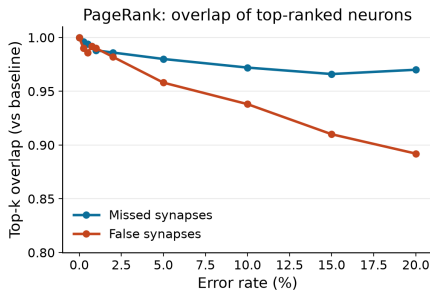
Reciprocity barely moves (−0.6%).

Giant SCC keeps 99.95% of vertices at 20% — wiring stays connected.

False Synapses — Graph Inflation and Structural Distortion

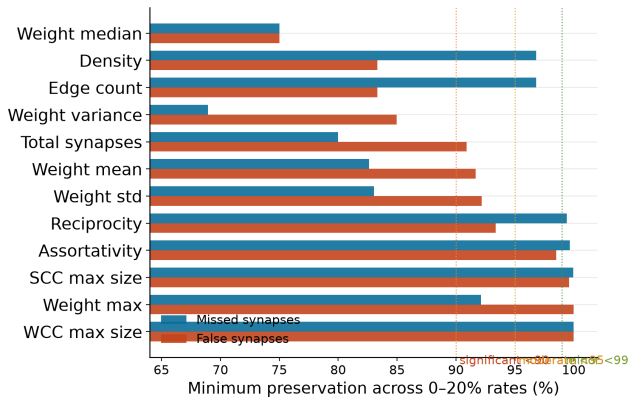


Degree distortion grows nonlinearly — KS distance $\approx 15\times$ larger than missed at 20%.



Hub ranking erodes: top- k overlap 0.89 at 20% (vs 0.97 for missed).
Added edges are weak (≈ 3 synapses vs. mean 5.9).

Comparison — False Synapses Are More Disruptive



Minimum preservation per metric, 0–20% rates (thresholds:
<90/ <95/ <99).

Metric @20%	Missed	False
Total synapses	−20.0%	+10.0%
Edges	−3.2%	+20.0%
Reciprocity	−0.6%	+7.1%
SCC giant	−0.05%	+0.4%
PageRank corr.	0.997	0.983
Top- <i>k</i> overlap	0.97	0.89

Same nominal rate —
larger structural effect.

Category-wise Preservation at 20% Error

Category	Metrics	Missed	False
Structural Topology	3	97.87%	88.89%
Synaptic Properties	7	83.10%	90.67%
Connectivity	2	99.97%	99.80%
Network Organization	2	99.53%	95.95%

Metrics = number of preservation metrics contributing to the category average.

At the 20% error rate — unweighted mean of member-metric preservation scores (calculation on next slide). Weight / assortativity values recomputed with the framework's current formula (see `correction.md`).

Most robust — global architecture

Node count, giant WCC/SCC (>99.6%), PageRank correlation (≈ 0.98 –1.0) and top- k overlap stay high under both models.

Most sensitive — synaptic layer

Synaptic statistics respond first under missed (weight variance 69%, median 75%, synapses 80%); edge count / density under false (83%); KS distorts 14–27 $\times$.

Trend shape

Primary quantities are *linear* in rate (synapse loss, edge gain); secondary effects (distances, reciprocity) are sublinear and accelerating.

How is Preservation Calculated?

Symmetric preservation ratio

Each graph metric is first converted into a preservation score using the symmetric ratio

$$\text{Preservation (\%)} = \frac{\min(\text{Baseline}, \text{Perturbed})}{\max(\text{Baseline}, \text{Perturbed})} \times 100$$

where *Baseline* = original graph metric and *Perturbed* = graph metric after reconstruction error.

Category preservation is then computed as the *unweighted arithmetic mean* of all preservation metrics belonging to that biological category.

Example — category = Average(member metrics)

Structural Topology = Average(Node Count, Edge Count, Density)

Synaptic Properties = Average(Total Synapses, Weight Mean, Weight Median, Weight Variance, Weight Standard

Deviation,

Weight Maximum, Weight Minimum)

Connectivity = Average(Largest WCC, Largest SCC)

Network Organization = Average(Reciprocity, Degree Assortativity)

Scientific Note: Rather than combining heterogeneous graph metrics into one overall preservation score, the report summarizes preservation separately for each biological category. This provides a more interpretable view of how reconstruction errors affect different aspects of the connectome while preserving the original metric-level calculations.

Reliability of the Results

- ▶ **5 seeded trials** per rate $\times$ 10 rates $\times$ 2 models (490 aggregated rows per model).
- ▶ **Tight confidence intervals:** total-synapse σ is only $\approx 0.01\%$ of the mean at 20%.
- ▶ **Deterministic perturbation:** false edges reproduce exactly ($\sigma = 0$); missed trials agree to within $\approx 0.01\%$ of the mean.
- ▶ **Monotone** primary trends (synapses, edges, KS) across all ten rates — no flips; higher-order metrics show only mild low-rate non-monotonicity.
- ▶ **Consistent status labels** (Preserved $\geq 99\%$ / Minor $\geq 95\%$ / Moderate $\geq 90\%$ / Significant $< 90\%$) across trials.

Spread @20% — σ / mean (rel.)

Metric	Missed	False
Total synapses	$\pm 0.009\%$	$\pm 0.005\%$
Edge count	$\pm 0.005\%$	$\pm 0.000\%$
SCC giant	$\pm 0.010\%$	$\pm 0.006\%$

Conclusions

Direct, measurable effects

Missed synapses remove synapses *exactly* in proportion to the rate (-20.0% at 20%); false synapses add edges *exactly* in proportion ($+20.0\%$). These are the most sensitive quantities measured.

Topology is highly robust

The giant connected component and node count survive to $>99\%$ even at 20% error — the coarse wiring plan is barely disturbed (PageRank correlation stays ≥ 0.98).

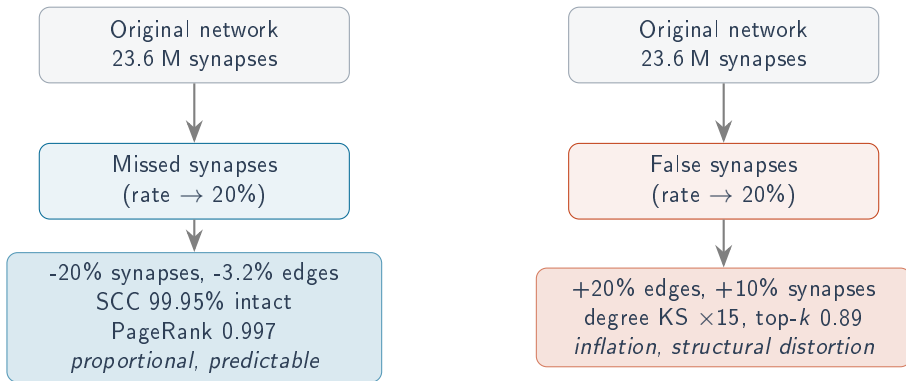
False synapses are the more disruptive model

At equal nominal rates they distort degree distributions $14\text{--}27\times$ more (in-degree KS), erode hub ranking (top- k 0.89 vs 0.97 for missed) and raise reciprocity by $+7\%$.

Error signatures are separable

Missed synapses *deflate* synaptic statistics; false synapses *inflate* the graph — weak edges (≈ 3 synapses) dilute weights without creating strong hubs.

Summary



Takeaway: reconstruction errors degrade synaptic statistics first — global wiring stays intact. False-positive errors are consistently more disruptive than missed synapses at the same rate.